

IDCM Axion & KK Tower

Date: 2026-07-18

Version: v1.0

Status: ✔ Framework confirmed

1. IDCM Axion

1.1 Origin

The IDCM axion arises from the antisymmetric tensor field B_2 (closed string moduli) on the CY_3 . On the $(36, 98)$ CY_3 , Hodge decomposition gives $h^{1,1} = 36$ axion fields.

1.2 Decay Constant and Mass

The axion decay constant f_a is locked at the GUT scale by the CY_3 volume:

$$f_a \approx \frac{M_P}{\sqrt{\text{Vol}(CY_3)}} \approx \frac{M_P}{\sqrt{1/\kappa^3}} \approx 3.0 \times 10^{16} \text{ GeV}$$
$$m_a = \frac{\Lambda_{\text{QCD}}^2}{f_a} \approx \frac{(0.18 \text{ GeV})^2}{3.0 \times 10^{16} \text{ GeV}} \approx 10^{-9} \text{ eV}$$

Parameter	IDCM Value	Typical Axion Range
f_a	$3.0 \times 10^{16} \text{ GeV}$	$10^{10} - 10^{17} \text{ GeV}$
m_a	$\sim 10^{-9} \text{ eV}$	$10^{-12} - 10^{-4} \text{ eV}$

1.3 Strong CP Problem

The IDCM axion dynamically relaxes the QCD θ parameter to zero:

$$\mathcal{L} \supset \frac{a}{f_a} G \tilde{G}, \quad \theta_{\text{QCD}} \rightarrow 0$$

No fine-tuning required. Strong CP is naturally solved in IDCM.

1.4 Axion Dark Matter

IDCM dark matter is NOT the axion, but the W-field KK mode (13.68 MeV). The axion's high f_a ($\sim$ GUT scale) would overclose the universe unless the initial misalignment angle $\theta_0 \ll 10^{-3}$ or inflation dilutes the abundance.

2. KK Excitations

2.1 KK Tower Structure

The W-field wrapped around S_w^1 generates a KK tower:

$$M_n = M_P \cdot e^{-n}, \quad n = 0, 1, 2, \dots, 42$$

The cutoff $N_h = 42$ comes from the IDCM computational limit.

KK Level n	Mass M_n	Physical Role
0	$M_P \approx 10^{19}$ GeV	Right-handed neutrino (zero mode)
1–5	$10^{17} - 10^{15}$ GeV	GUT scale particles
6–10	$3 \times 10^{16} - 5 \times 10^{14}$ GeV	Intermediate scale
20–25	$2 \times 10^{10} - 10^8$ GeV	Beyond collider reach
36	~ 2.8 TeV	Detectable at FCC-hh
37	~ 1.0 TeV	HL-LHC limit
40	~ 140 GeV	Sub-Higgs scale
42	13.68 MeV	Dark matter particle 

2.2 Experimental Detectability

- **n=36-37 (1-3 TeV):** Potential KK graviton resonance at future colliders (FCC-hh)
- **n=42 (13.68 MeV):** Matches observed DM density (non-thermal W-field vacuum condensation)
- **Low-level KK:** Gravitational coupling too weak; only indirectly detectable via gravitational waves (LISA)

3. Summary

IDCM predicts two types of beyond-SM particles:

Particle	Mass	Role	Detectability
W-field KK mode $n = 42$	13.68 MeV	 Dark matter	Indirect (rare decays)
QCD axion	$\sim 10^{-9}$ eV	 Strong CP solution	CASPEr/ABRACADABRA
KK graviton $n = 36$	~ 2.8 TeV	 Collider signal	FCC-hh